ENVS 193DS Spring 2026 Final

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Setup

```
library(tidyverse)
library(here)
library(janitor)
library(MuMIn)
library(DHARMa)
library(ggeffects)
nests <- read_csv(here("data","nest_data_final.csv"))
bike <- read_csv(here("data","ENVS-193DS-Personal-Data-Project-cont.csv"))
```

Problem 1. Research writing

a. Transparent statistical methods

In part 1, they used a logistic regression to predict the probability that American Avocets would use microhabitats based on distance to water edge.

In part 2, they used a chi-square test to determine if there is a relationship between bird species and habitat type.

b. Suggestions for rewriting

American Avocets were more likely to use some microhabitats more than others based on distance to water edge. The habitat's distance to water edge seems to have an influence on probability that American Avocets would use it.

Habitat use differed between bird species (American Avocet, Forster's Tern, Black-necked Stilt), based on its type (managed pond, non-tidal marsh, salt pond, tidal marsh, tidal mudflat). The bird species were not evenly distributed among these habitat types.

c. Further analysis

Another variable that could influence the probability of American Avocet microhabitat use is distance to nearest road. This would be a continuous variable measured in meters to the nearest road that regular cars drive on. The noise and air pollution from the cars might disrupt and discourage the birds from using the microhabitat, meaning probability of microhabitat use would most likely increase the further the habitat is from a road.

Problem 2. Data analysis

a. Clean and wrangle

```
# creating new object from data
nests_clean <- nests |>
  # selecting columns of interest
  select(height_cm, case_control, ant_species) |>
  # filtering for species of interest
  filter(ant_species == c("RRB", "AB", "BBR")) |>
  # adding new column with ant scientific name
  mutate(scientific_name = recode_values(ant_species,
    "AB" ~ "Crematogaster sjostedti",
    "RRB" ~ "Crematogaster mimosae",
    "BBR" ~ "Crematogaster nigriceps"
  )) |>
  # reordering levels of ant species
  mutate(ant_species = as_factor(ant_species),
         ant_species = fct_relevel(ant_species,
                                   "AB",
                                   "RRB",
                                   "BBR"))
```

```
# displaying 10 random rows from table
slice_sample(nests_clean, n = 10)
```

```
# A tibble: 10 x 4
   height_cm case_control ant_species scientific_name
       <dbl>        <dbl> <fct>       <chr>
 1      169             0 RRB         Crematogaster mimosae
 2      360.            0 RRB         Crematogaster mimosae
 3      119             0 RRB         Crematogaster mimosae
 4      312             0 RRB         Crematogaster mimosae
 5      191             1 RRB         Crematogaster mimosae
 6      524.            1 RRB         Crematogaster mimosae
 7      118             0 RRB         Crematogaster mimosae
 8      260.            0 RRB         Crematogaster mimosae
 9      157             0 RRB         Crematogaster mimosae
10      114.            0 RRB         Crematogaster mimosae
```

b. Response variable

In the `case_control` column, which is the response variable, the 1s represent a tree that has a bird nest in it while the 0s represent trees that did not have a bird nest in it.

c. Purpose of study

Tree height is a continuous variable measured in cm. As tree height increases, the probability of nest occurrence increases because the canopy is denser higher up and provides more cover for the bird nest. This is described in the second paragraph in the Discussion section.

Acacia ant species is a categorical variable, where its presence on a host tree is denoted by a code unique to its species. Based on biological relationships, trees with *C. nigriceps* and *C. mimosae* increase the probability of nest occurrence much more than trees with *C. sjostedti* because they defend their trees from herbivores and bird predators much more aggressively. This is described in the first and second paragraphs in the Discussion section.

d. Table of models

2 models total:

Model number	Tree Height	Ant Species	Model Description
1	X	X	all predictors (no interaction term)
2	X	X	all predictors (with interaction term)

e. Run the models

```
# model 1: no interaction
nests_mod1 <- glm(case_control ~ height_cm + ant_species,
                  data = nests_clean,
                  family = "binomial")

#model 2: with interaction
nests_mod2 <- glm(case_control ~ height_cm * ant_species,
                   data = nests_clean,
                  family = "binomial")
```

f. Select the best model

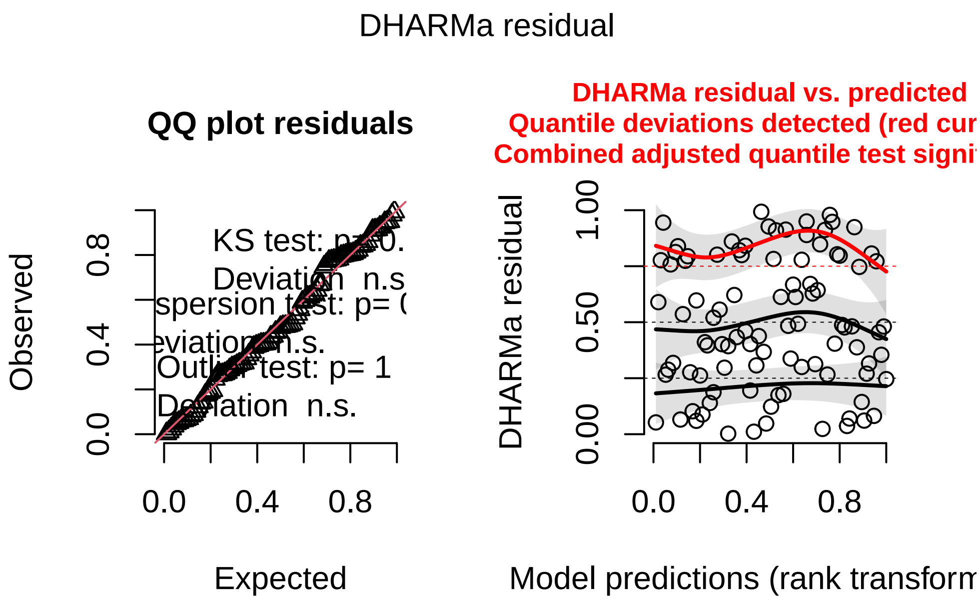
```
AICc(
  nests_mod1,
  nests_mod2
) |>
  arrange(AICc)
```

```
           df     AICc
nests_mod2  6 86.95550
nests_mod1  4 92.84684
```

The best model as determined by Akaike's Information Criterion (AIC) was the probability of nesting in trees as an interactive effect of height and ant species.

g. Check the diagnostics

```
plot(
  simulateResiduals(nests_mod2)
)
```

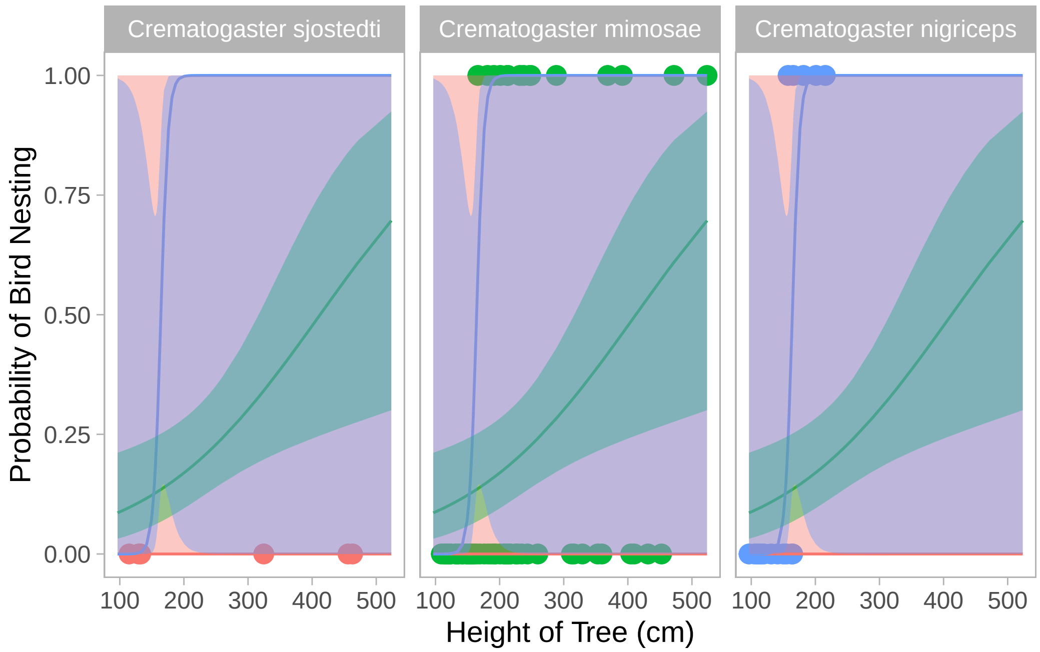


h. Visualize the model predictions

```
# creating a new object
nests_mod2_preds <- ggpredict(
  nests_mod2, # model
  terms = c("height_cm [all]", "ant_species") # predictor variable
)
```

```
# base layer: ggplot
ggplot(data = nests_clean,
       # x-axis: height
       # y-axis: probability of bird nesting
       aes(x = height_cm,
           y = case_control)) +
  # first layer: points to represent individual observations
  geom_point(aes(color = ant_species),
             size = 3,
             alpha = 1) +
  # second layer: model prediction as a line
  geom_line(data = nests_mod2_preds,
            aes(x = x,
```

```r
              y = predicted,
              color = group)) +
# third layer: 95% CI from model prediction as a line
geom_ribbon(data = nests_mod2_preds,
            aes(x = x,
                y = predicted,
                ymin = conf.low,
                ymax = conf.high,
                fill = group),
            alpha = 0.4) +
# separating into 3 graphs for each ant species
facet_wrap(~ ant_species,
           # relabeling each graph
           labeller = labeller(
             ant_species = c(AB = "Crematogaster sjostedti",
                             RRB = "Crematogaster mimosae",
                             BBR = "Crematogaster nigriceps"))) +
# changing theme
theme_light() +
# removing legend
theme(legend.position = "none") +
# relabeling x and y-axes
labs(x = "Height of Tree (cm)",
     y = "Probability of Bird Nesting") +
# removing gridlines
theme(panel.grid = element_blank())
```



i. Wite a caption for your figure

Figure 2. Probability of bird nesting based on tree height and ant species This figure shows the probability of bird nesting based on tree height and what ant species the tree is hosting. Birds primarily nested in trees with *C. mimosae* and *C. nigriceps*. Data citation: Lujan, E., Nielsen, R., Short, Z., Wicks, S., Nderitu Watetu, W., Malingati Khasoha, L., Palmer, T. M., Goheen, J. R., & Alston, J. M. (2023). Data, code, and metadata for: Symbiotic acacia ants drive nesting behavior by birds in an African savanna [Data set]. Zenodo. https://doi.org/10.5281/zenodo.8373322

j. Calculate model predictions

```
# predicting nesting probability when tree height is 600cm
ggpredict(nests_mod2,
          terms = c("height_cm [600]",
                    "ant_species"))
```

```
# Predicted probabilities of case_control

ant_species: AB
```

```
height_cm | Predicted |     95% CI
----------------------------------
      600 |      0.00 | 0.00, 1.00

ant_species: RRB

height_cm | Predicted |     95% CI
----------------------------------
      600 |      0.80 | 0.34, 0.97

ant_species: BBR

height_cm | Predicted |     95% CI
----------------------------------
      600 |      1.00 | 0.00, 1.00
```

k. Interpret your results

Tree height and ant species presence interact to predict probability of bird nesting in acacia trees. As tree height increases, the probability of bird nesting increases. Additionally, the ant species *Crematogaster mimosae* and *Crematogaster nigriceps* had a great effect on nesting probability than *Crematogaster sjostedti*. At tree height 600cm, bird nesting probability is 0 (95% CI: [0.00, 1.00]) for *Crematogaster sjostedti*, while it is 0.80 (95% CI: [0.34, 0.97]) for *Crematogaster mimosae* and 1.00 (95% CI: [0.00, 1.00]) for *Crematogaster nigriceps* respectively.

This relationship aligns with biological trends. Bird nesting is more successful in higher trees given it is further away from grazing mammals and predators. The two ant species *Crematogaster mimosae* and *Crematogaster nigriceps* are more aggressive in protecting their host trees, allowing for more successful bird nesting in their presence.

Problem 3. Affective and exploratory visualizations

a. Comparing visualizations

My exploratory visualization from Homework 2 and affective visualization from Homework 3 ended up somewhat similar with some key differences. My visualization from Homework 2 did not have many data points yet and I made it a line graph, although this did not make much sense in hindsight. In comparison, my affective visualization from Homework 3 contained a lot more data points, although the upward trend first seen in Homework 2's visualization was the same. In general, as duration of the day's activities increased, my time spent biking also increased. I ended up representing this upward trend as a mountain slope in my final affective visualization. The main feedback I received during week 9 workshop was to clarify portions of my artist statement, which I did for the final presentation.

b. Designing an analysis

Based on my response variable and predictor variable, I think an appropriate analysis would be a linear regression. My initial research question was about how the total duration of the day's activities (classes, work, etc.), measured in hours, affected my total daily time spent biking, measured in minutes. I initially hypothesized that a busier day would mean more time spent biking as I traveled from home to campus to work, so a linear regression would be able to predict this relationship.

Both my response variable, total daily time spent biking, and my predictor variable, total duration of the day's activities, are continuous variables. A linear relationship between these two variables could be feasible as well, so a linear regression was appropriate for this analysis.

c. Check any assumptions and run your analysis

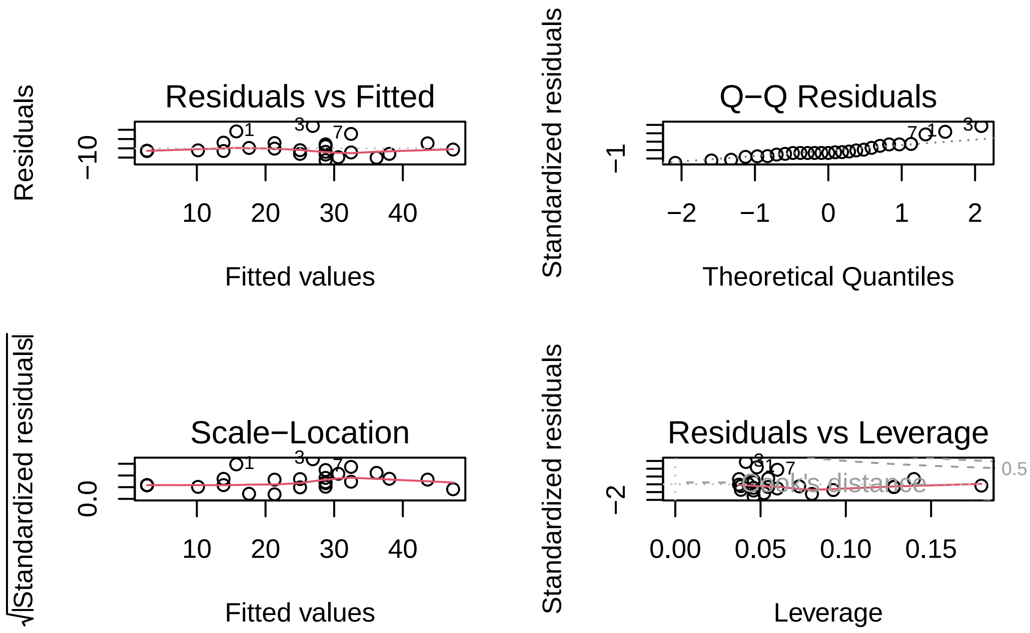
Model fitting

```r
# creating new object
bike_clean <- bike |>
  # cleaning column names to remove spaces
  clean_names()
```

```r
bike_model <- lm(
  duration_of_bike_ride_minutes ~ duration_of_days_activities_hours, # formula: duration of b
  # data frame: bike
  data = bike_clean
)
```

Diagnostics

```r
# display plots in a 2x2 grid
par(mfrow = c(2, 2))
# display diagnostic plots for model
plot(bike_model)
```



- Residuals vs Fitted and Scale-Location: points are roughly distributed on both sides of reference line, so residuals have equal variance
- Q-Q Residuals: points roughly follow QQ reference line, so residuals are normally distributed
- Residuals vs Leverage: no obvious outliers outside of dotted lines that would affect model

Model summary

```r
# display summary of linear model
summary(bike_model)
```

```
Call:
lm(formula = duration_of_bike_ride_minutes ~ duration_of_days_activities_hours,
    data = bike_clean)

Residuals:
    Min      1Q  Median      3Q     Max
-12.747  -4.104  -2.744   3.253  24.111
```

```
Coefficients:
                                   Estimate Std. Error t value Pr(>|t|)
(Intercept)                          2.7436     3.0762   0.892    0.381
duration_of_days_activities_hours   3.7147     0.4864   7.637 5.42e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8.591 on 25 degrees of freedom
Multiple R-squared:    0.7, Adjusted R-squared:  0.688
F-statistic: 58.33 on 1 and 25 DF,  p-value: 5.416e-08
```

Generating predictions

```r
# creating a new object
bike_preds <- ggpredict(
  bike_model, # model
  terms = "duration_of_days_activities_hours" # predictor variable
)
# display predictions
bike_preds
```

```
# Predicted values of duration_of_bike_ride_minutes

duration_of_days_activities_hours | Predicted |       95% CI
------------------------------------------------------------
                             0.00 |      2.74 | -3.59,  9.08
                             3.00 |     13.89 |  9.76, 18.02
                             3.50 |     15.75 | 11.88, 19.61
                             5.00 |     21.32 | 17.90, 24.74
                             6.50 |     26.89 | 23.29, 30.49
                             7.50 |     30.60 | 26.57, 34.64
                             9.00 |     36.18 | 31.17, 41.18
                            12.00 |     47.32 | 39.82, 54.82
```

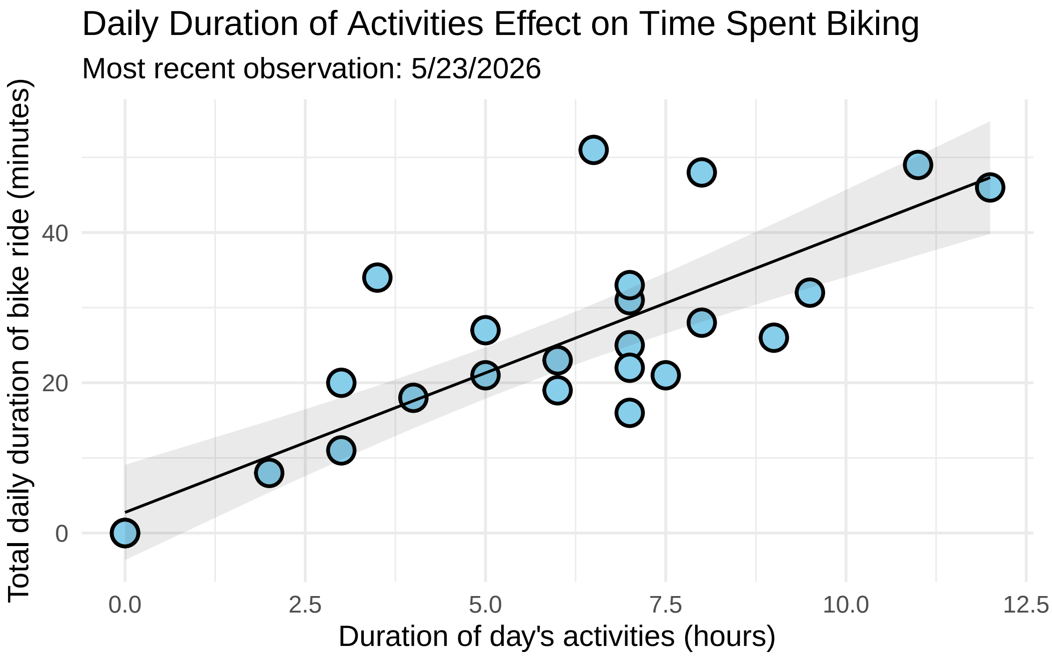
```r
ggpredict(
  bike_model, # model
  terms = "duration_of_days_activities_hours[10]" # predicting bike ride duration when activi
)
```

```
# Predicted values of duration_of_bike_ride_minutes
```

```
duration_of_days_activities_hours | Predicted |       95% CI
------------------------------------------------------------
                               10 |     39.89 | 34.11, 45.67
```

d. Create a visualization

```
# base layer: ggplot
ggplot(data = bike_clean,
       # x-axis: duration of day's activities
       # y-axis: duration of bike ride
       aes(x = duration_of_days_activities_hours,
           y = duration_of_bike_ride_minutes)) +
  # first layer: points to represent individual days
  geom_point(size = 4,
             stroke = 1,
             fill = "skyblue",
             shape = 21) +
  # second layer: ribbon representing confidence interval
  geom_ribbon(data = bike_preds,
              aes(x = x,
                  y = predicted,
                  ymin = conf.low,
                  ymax = conf.high),
              alpha = 0.1) +
  # third layer: line representing model predictions
  geom_line(data = bike_preds,
            aes(x = x,
                y = predicted)) +
  # relabeling axes, adding title and subtitle
  labs(x = "Duration of day's activities (hours)",
       y = "Total daily duration of bike ride (minutes)",
       title = "Daily Duration of Activities Effect on Time Spent Biking",
       subtitle = "Most recent observation: 5/23/2026") +
  # changing theme
  theme_minimal()
```



e. Write a caption

Figure 2. Daily Duration of Activities Effect on Time Spent Biking As duration of day's activities increases, daily total time spent biking also increases. Individual daily observations are shown as blue dots on the graph as well as a 95% confidence interval.

f. Write about your results

As the duration of my day's activities (classes, work, meetings, etc,), my daily total time spent biking also increases. When duration of activities = 10 hours, the model predicts total time spent biking that day to be 39.89 minutes [95% CI: 34.11, 45.67].